Evaluations of News Source Strategies to Disseminate Fake News in X

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Abstract

False-news sources can alter their presentation without changing publication veracity. This study evaluates such strategies on X using a survey-informed agent-based model with 500 synthetic users and four source objects. A staged design comprises 6,690 seeded runs across 423 conditions spanning imitation strategy, campaign timing, direct reach, recommendation policy and parameter and structural uncertainty. Combined imitation increased experimental-source decision share by 0.471 percentage points (pp) from a 1.328% control mean, while its 0.157 pp false-decision-share effect was imprecisely estimated. Short campaigns increased source share while active and produced a post-removal reversal that subsequently attenuated. At 14.7% direct reach, the combined false-decision-share effect was 0.149 pp under immediate perfect labels, compared with 9.097 pp under fast imperfect labels and 15.131 pp under delayed imperfect labels. Morris screening identified memory half-life as the largest influence on the combined strategy effect. Audience gains remained positive across all activity and topology cells, whereas false-exposure robustness depended on strategy and structure. The model therefore distinguishes source success from false exposure and shows that reach governance depends critically on recommendation accuracy and timeliness. Results are comparative model experiments, not predictions of X.

Keywords: agent-based modelling; false news; misinformation; news sources; social media; sensitivity analysis

1 Introduction

Fake news can impose costs well beyond an isolated inaccurate belief. In democratic settings, disinformation can distort public debate, intensify polarisation, and erode trust in elections, institutions, and journalism; the OECD therefore treats information integrity as an enabling condition for democratic participation and accountable government (OECD, 2024). Public-health decisions provide a second concrete example. In a randomised experiment in the United Kingdom and the United States, exposure to COVID-19 vaccine misinformation reduced the share of respondents who said they would definitely accept vaccination by 6.2 and 6.4 percentage points, respectively (Loomba et al., 2021). These examples do not imply that every false publication changes behaviour or determines an election. They show why the reach and repetition of deceptive content can create societal consequences that justify careful, rights-aware intervention.

Online social networks allow news sources to reach large audiences with low publication and redistribution costs. The same infrastructure supports legitimate journalism, uncertain information, deliberately deceptive content, and mixtures of these forms. False news has been shown to diffuse farther and faster than true news in large Twitter cascades (Vosoughi et al., 2018), while exposure and sharing are concentrated among subsets of users (Grinberg et al., 2019; Guess et al., 2019). These findings motivate interventions, but they do not by themselves reveal how a malicious source may adapt its observable presentation when platforms or users become more selective.

This paper focuses on strategies controlled by a news source that repeatedly disseminates false publications. Such a source may imitate attributes associated with established media: emotional engagement, political or social proximity, apparent information quality, or source credibility. We distinguish these perceived attributes from the objective probability that a publication is false. This separation is essential. A camouflage strategy can make a source more attractive without making its output more accurate. We use *false news* for factually false news-like publications, *misinformation* without assuming intent, and *disinformation* only when deliberate deception is established. Because the model specifies behaviour rather than an internal intention state, "malicious source" is shorthand for the experimental source and not an empirical diagnosis of motive.

Observed social-media data are indispensable but make controlled counterfactuals difficult. Agent-based modelling (ABM) complements observation by representing users with heterogeneous states and network positions, repeated decisions and feedback, and by changing one policy or strategy while holding the remaining design constant (Macal and North, 2010; Grimm et al., 2020). Our model therefore serves as a computational laboratory for source strategies and mitigation policies; it is not intended to forecast X at the individual-user level.

We address four research questions:

RQ1. How do source-imitation strategies affect malicious-source reach and fake-news dissemination? Platforms and users observe many source characteristics, but it is unclear which groups of characteristics are most exploitable or whether a source gaining audience necessarily increases network-wide false-news exposure. RQ1 contributes a controlled ranking of individual and combined imitation strategies while holding objective fake-news probability fixed.

RQ2. How do campaign timing and duration affect efficacy and persistence? Deceptive presentation may be adopted early, late, or temporarily, while detection and intervention occur after different delays. A full-run average confounds start time with exposure duration, so RQ2 contributes matched treatment and post-removal windows that distinguish opportunity during a campaign from persistence after it ends.

RQ3. How effectively does limiting experimental-source reach mitigate dissemination, and how do recommendation policies modify this relationship? Reach and recommendation are platform-controllable mechanisms, but interpersonal diffusion could complement or bypass direct visibility restrictions. An immediate and perfectly truth-aware mechanism is an informative boundary, not a realistic platform description. RQ3 therefore contributes a joint assessment of graded reach limits, discovery-only recommendation, asymmetric flagging, delayed and imperfect labels, and recommendation magnitude.

RQ4. Are strategy conclusions robust to parameter and structural uncertainty? Memory persistence and contact counts are uncertain behavioural assumptions rather than direct policy levers, and conclusions may also depend on source behaviour, choice scaling, user activity and network construction. RQ4 contributes local, global and structural sensitivity analyses that distinguish stable conclusions from effects requiring stronger behavioural qualifications.

The contribution is threefold. First, we provide a modular ABM in which named source strategies selectively copy observable attributes while publication veracity remains unchanged. Second, we use common random-number seeds in a staged computational experiment to compare strategy, timing, reach and recommendation policies and to screen parameter and structural uncertainty. Third, we distinguish four outcomes that policy discussions can otherwise conflate: false reposts per agent-period, false reposts among actual decisions, experimental-source share among decisions, and participation.

The remainder of the paper is organised as follows. Section 2 reviews fake-news dissemination, ABM, and related simulation applications. Section 3 describes the model and experimental design using the ODD protocol. Section 4 reports the four research-question analyses. Section 5 documents software verification and conceptual and behavioural validation. Section 6 develops societal and policy implications while delimiting the claims supported by the experiments. Section 7 concludes and identifies the next validation steps.

2 Background: fake news and simulation models for analysing its dissemination

This section establishes the conceptual and methodological foundations of the study. It first explains why fake-news dissemination is mediated by source identity and social interaction, then introduces ABM and the ODD reporting protocol, and finally positions the proposed model relative to existing misinformation simulations.

2.1 Fake news and source-mediated dissemination

Fake news is not merely erroneous information; it is fabricated content presented in a news-like form and connected to incentives, attention, and institutional trust (Lazer et al., 2018; Allcott and Gentzkow, 2017). Social platforms alter the route by which people encounter this content. Users select from sources, observe peers, and react to signals such as credibility, emotional tone, familiarity, and proximity. Homophilous communities can reinforce selective exposure (Del Vicario et al., 2016), and social influence can change evaluations independently of intrinsic quality (Muchnik et al., 2013).

Source identity matters because users often lack the time or information needed to verify each claim. Aggregated judgments of source quality can support misinformation interventions (Pennycook and Rand, 2019), whereas low-credibility content may receive coordinated amplification from automated accounts (Shao et al., 2018). These mechanisms make source-level policies—visibility limits, recommendation rules, labels, and sanctions—plausible levers. They also create adaptation incentives: a malicious source may modify visible attributes to resemble a trusted source.

2.2 Agent-based models as simulation models

ABMs describe system outcomes as the emergent result of local decisions and interactions. They are useful when agents differ, remember past events, learn from contacts, and respond to interventions over time. In social-network research, ABMs have evaluated word-of-mouth (WOM) campaigns and participant selection (Leger et al., 2016; Plaza et al., 2019), message repetition on Twitter (López et al., 2023), and extensible architectures for social-network simulation (Leger et al., 2026b). These applications illustrate how an executable mechanism can connect individual exposure and choice to population-level diffusion while retaining a controlled experimental design.

A credible ABM must make its entities, processes, scheduling, stochasticity, inputs, and outputs explicit. We follow the ODD protocol (Grimm et al., 2020), which organises a model description into Overview, Design concepts, and Details. Figure 1 summarises how those components move from the scientific purpose and system boundary to behavioural assumptions and executable submodels. ODD supports inspection and replication, but does not itself establish that a model is valid for every use.

Simulation evidence has a different status from observational or experimental evidence about people. A simulation supports conditional statements: given the encoded mechanisms and parameter ranges, a strategy produces a measured change. Verification asks whether the software implements that specification; validation

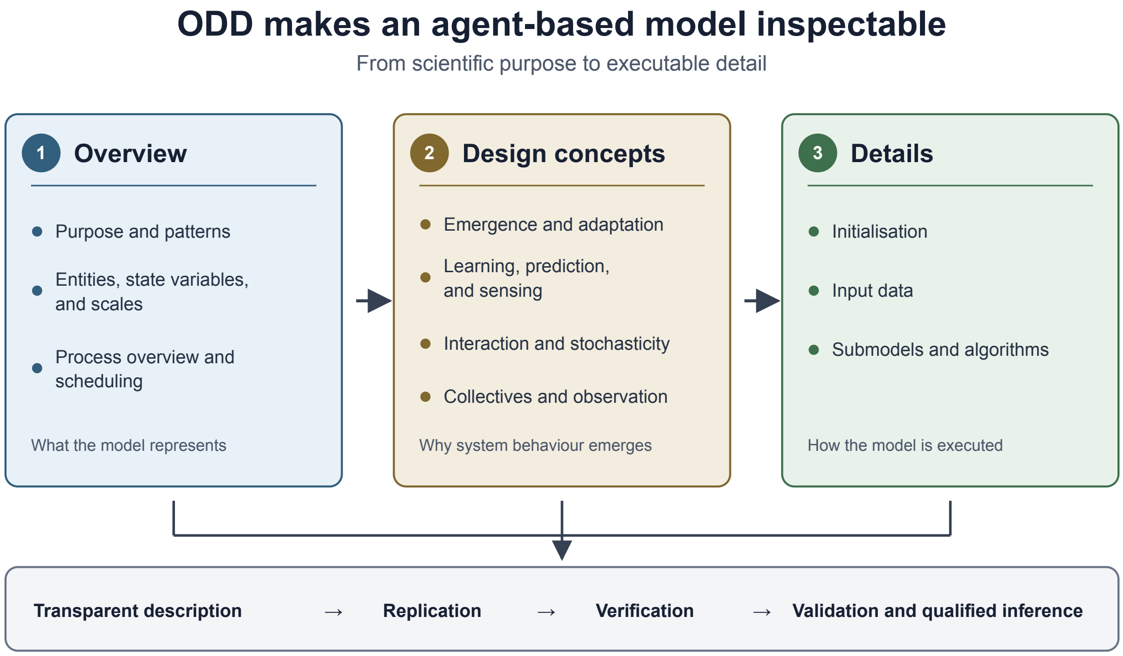

Figure 1: Original synthesis of the ODD protocol for documenting an agent-based model. The three descriptive components support replication, software verification, and qualified validation.

asks whether the representation is sufficiently credible for its intended purpose. Neither process turns an exploratory model into a direct prediction of platform behaviour.

2.3 Simulation models of misinformation

Misinformation models range from epidemic analogies to network, opinion-dynamics, and cognitively richer agent systems. Table 1 compares representative models by their behavioural focus, empirical input, intervention and validation. Competing-rumour models represent misinformation and fact-checking as diffusion processes (Sulis and Tambuscio, 2020); game-theoretic models examine deception and cooperation (Kopp et al., 2018); curation models connect belief, recommendation objectives and empirically observed Twitter data (Gausen et al., 2023); and bot models study how network position changes indirect influence (Keijzer and Mäs, 2021). Recent work also reconstructs language and network histories through digital cloning (Puri et al., 2024), examines moderation across platforms (Murdock et al., 2024), and evaluates governance in decentralised systems (Malgonde et al., 2025).

Our model occupies a complementary niche. It does not reconstruct a particular X network or classify text. Instead, it makes a source—rather than a message or belief state—the adaptive experimental actor, keeps objective false-publication behaviour separate from perceived attributes, and evaluates selective and combined source imitation. This makes it suitable for controlled scenario analysis and for distinguishing source-level success from system-level false-news exposure. Its novelty is therefore the experimental treatment of source presentation and platform-controllable reach, not higher-fidelity reconstruction of X.

Table 1: Positioning relative to representative misinformation simulation models. "Empirical" identifies inputs or validation data reported by each study; it does not imply predictive validity.

Study	Main mechanism	Network/input	Intervention	Distinctive analytical focus
Sulis and Tambuscio (2020)	Competing misinformation and debunking states	Configurable synthetic networks	Fact-checking probability	Diffusion and forgetting
Kopp et al. (2018)	Evolutionary deception and cooperation	Random interactions; empirical pattern comparison	Behavioural costs	Deception types and cooperation
Gausen et al. (2023)	Beliefs and newsfeed curation	Parameterisation and case comparisons using Twitter data	Curation objectives	Algorithmic misinformation–polarisation trade-off
Keijzer and Mäs (2021)	Cultural influence by bots and humans	Synthetic network structures	Bot activity and connectivity	Direct versus indirect bot influence
Puri et al. (2024)	Language-sensitive digital clones	Observed network and message histories	Counterfactual content	Reproduction of a particular online system
This study	Repeated source choice from accumulated evidence	Survey-informed source perceptions; synthetic directed networks	Source imitation, reach and recommendation policy	Adaptive source presentation with objective false-publication probability held separate

3 Methodology

This section specifies the model using the ODD protocol, defines the outcomes and experimental contrasts, and documents the computational workflow. Appendix A provides the complete executable-level ODD specification and parameter inventory; the present section concentrates on decisions needed to interpret the experiments.

3.1 Overview: purpose, entities, state variables and scale

The model is a comparative laboratory for strategies by which an experimental false-news source changes observable attributes and for platform policies that alter its dissemination. A run contains 500 user agents and four individual source objects representing categories: traditional media, an emerging or unfamiliar online outlet ("unknown media"), the experimental false-news source, and a mixed source that combines comparatively credible and non-credible characteristics. The model does not contain multiple outlets within each category.

Each run lasts 400 discrete decision periods. A period is an ordering device and has no fixed mapping to minutes, days or numbers of posts. The horizon permits transient, campaign and post-removal windows to be observed while remaining computationally tractable. There is no discarded warm-up period. Initial evidence is an explicit part of the model and early-horizon outcomes are reported to expose, rather than conceal, transient effects.

The four source objects carry perceived attribute distributions and an objective false-publication probability. Perceived attributes cover evoked emotion, simple language, political, geographic and social proximity, sensationalism, information quality, entertainment, audiovisual content, links, hashtags and source credibility. The respective false-publication probabilities are 0.092, 0.206, 0.667 and 0.416. These values reproduce the low-credibility probabilities in the survey-informed workbook and are modelling inputs, not observed estimates of each category's factual error rate. A strategy may copy perceived attributes but never copies objective publication behaviour.

All 500 agents share one vector of survey-derived mean preference weights. Claims of preference heterogeneity would therefore be incorrect. Agents become state-heterogeneous through initial source visibility, directed contacts, stochastic attribute histories, publication exposure and source choices. The structural sensitivity experiment additionally varies activity and network construction. This distinction between homogeneous preference parameters and heterogeneous realised states is maintained throughout the paper.

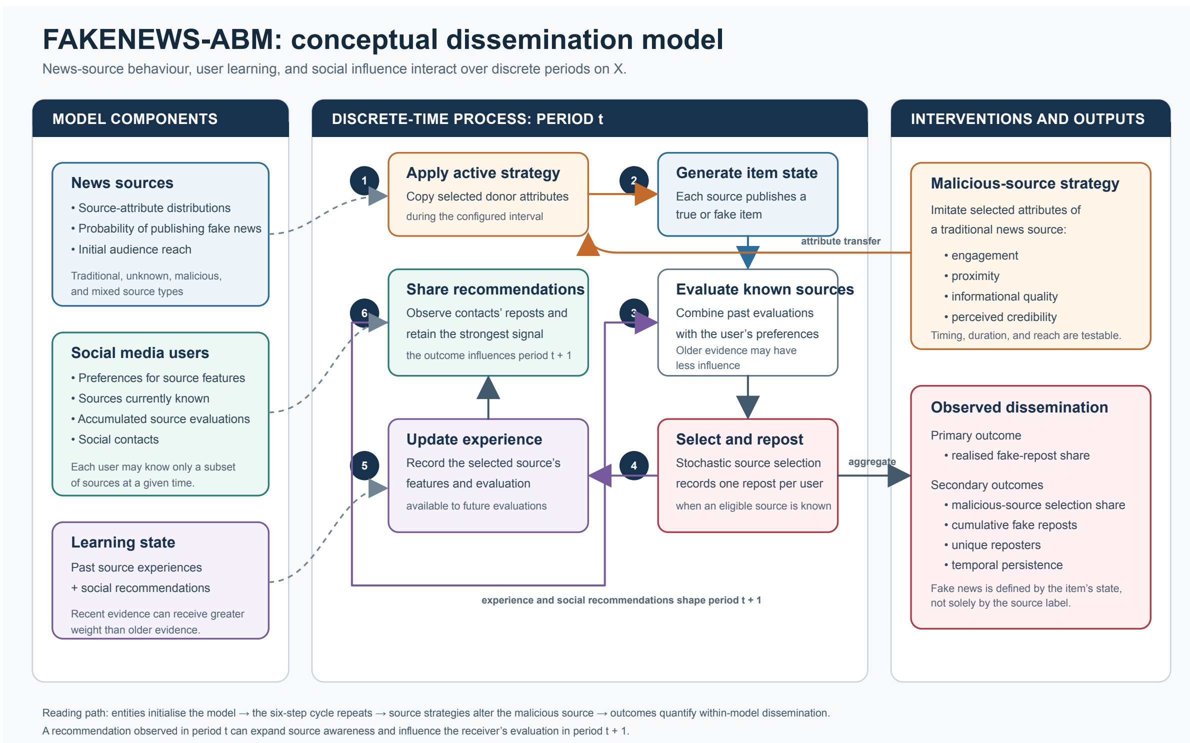

Figure 2: Conceptual structure of the ABM. One source-period publication state is generated before user decisions and is shared by all users who select that source in the period.

3.2 Process overview and scheduling

At reset, the simulator constructs sources and users, samples initial source visibility, builds a directed contact network and creates deterministic endorsement events at period -1. For each initially known source and attribute, initialization chooses the more probable attribute level and converts it to a signed endorsement. This provides a defined first-period choice distribution; it represents prior familiarity and is not an empirically observed interaction history.

During period t: (1) scheduled strategies are activated or restored; (2) each source draws one binary publication state $Z_{st} \sim \text{Bernoulli}(p_s)$; (3) pending recommendation evidence is delivered; (4) active users evaluate known sources and select at most one; (5) selection, false-publication and unique-reposter outcomes are recorded; and (6) each user observes the strongest current recommendation among directed contacts. Every user who selects source s in period t reposts the same source-period state Z_{st}. In the principal experiments, an agent with a known source makes exactly one decision and voluntary abstention is absent.

Reach restriction can leave an agent with no known source, and the structural experiment introduces an explicit activity probability.

For an endorsement event e with signed value v_e, source score is

$$E_{is}(t) = \sum_{e \in \mathcal{M}_{is}(t)} \operatorname{sgn}(v_e) B^{|v_e|} w_e(t), \tag{1}$$

where B is the choice-function base and $\mathcal{M}_{is}(t)$ is the retained memory. With a hard window of M periods, retained events receive constant weight. With infinite memory and exponential decay,

$$w_e(t) = 2^{-(t-\max\{1,t_e\})/h}, \tag{2}$$

where h is the half-life. The principal $h = 0.33$ is a legacy scenario anchor, not an empirical estimate: an event retains approximately 12.2% of its transformed contribution after one full period. Its extremity motivates the wider half-life and hard-window sensitivity experiments; the two memory representations are not claimed to be behaviourally equivalent.

Here i indexes users, s sources, e endorsement events, t_e the event period, v_e the signed event value, and $\operatorname{sgn}(v_e) \in \{-1, 0, 1\}$ its sign. The term $\max\{1, t_e\}$ assigns the deterministic initialization events dated -1 age zero at the first decision; ordinary events retain their recorded period.

If every $E_{is}(t) \geq 0$ and their sum is positive, source-choice probability is $E_{is}/\sum_r E_{ir}$. Otherwise the implementation shifts every candidate by the minimum score and adds one:

$$P_i(s,t) = \frac{E_{is}(t) - \min_r E_{ir}(t) + 1}{\sum_q [E_{iq}(t) - \min_r E_{ir}(t) + 1]}. \tag{3}$$

This makes every known source selectable even when evaluations are non-positive.

The baseline network is directed with fixed outdegree $\lfloor \texttt{CONTACTS} \times \texttt{FRIENDS} \rfloor = 5$ and no self-links or duplicate destinations. Five is a parsimonious scenario anchor, not a documented empirical mean. It is consistent only with the broader finding that active interaction networks on Twitter are much sparser than declared follower links (Huberman et al., 2009); RQ4 therefore varies the count from 3 to 10. Indegree is unconstrained and variable. For each source selected by a contact, only the largest contact evaluation is retained; this prevents repeated recommendations of one source from being summed mechanically, but may privilege an extreme contact evaluation. The source with the largest retained evaluation is recommended, with source-identifier iteration order resolving exact ties. The diagnostic re-executions report duplicate recommendations and ties so this deterministic rule is auditable. The sender's evaluation selects the recommendation, while the receiver's word-of-mouth (WOM) preference determines endorsement magnitude. A received recommendation makes a previously unknown source visible before label coverage is sampled.

Recommendation evidence is parameterised by coverage c, sensitivity s_e, specificity s_p, delay d and receiver scale λ. Coverage is sampled separately for the one selected sender–receiver recommendation in every receiver-period. When a label is available, its observed state is sampled using s_e and s_p; the configured false/true policy then penalises, rewards or ignores the recommendation. Because all agents share the same preference vector, the non-zero contribution is $g\lambda W$ and is delivered at $t + 1 + d$. Table 2 gives the complete named-policy mapping. The original oracle boundary has $(c, s_e, s_p, d) = (1, 1, 1, 0)$; the revision adds discovery-only, asymmetric flagging, delayed and imperfect policies and $\lambda \in \{0.25, 0.5, 1\}$.

3.3 Input parameterisation and scope of empirical claims

The input workbook was created from aggregate responses from an online survey of Chilean X users. The delivery file contains 510 complete records; the model population of 500 agents is a computational design choice, not the survey sample size. The workbook supplies category-level source-attribute distributions,

Table 2: Complete configuration of named RQ3 recommendation policies. Coverage is sampled for each selected receiver–recommendation event. Discovery occurs before coverage; therefore an uncovered or ignored recommendation can still reveal a source.

Policy	WOM	g_F	g_T	c	s_e	s_p	d	λ
Disabled	off	–	–	–	–	–	–	.50
Discovery only	on	0	0	1.00	1.00	1.00	0	.50
False-only flag	on	−1	0	1.00	1.00	1.00	0	.50
Perfect immediate (oracle)	on	−1	+1	1.00	1.00	1.00	0	.50
Imperfect fast	on	−1	0	.80	.90	.95	5	.50
Imperfect delayed	on	−1	0	.50	.75	.90	25	.50

g_F and g_T are the effects assigned to an observed false or true label: -1 penalises, 0 ignores and $+1$ rewards. Sensitivity s_e is $P(\text{observed false} \mid \text{false})$; specificity s_p is $P(\text{observed true} \mid \text{true})$. Receiver-scale experiments additionally use $\lambda \in \{.25, 1.00\}$ with the imperfect-fast policy.

mean user weights and initial reach values of 81.2%, 46.7%, 14.7% and 17.5%. We call this survey-informed parameterisation rather than calibration: respondent-level vectors and their correlations are not represented, and no observed X cascade was used to fit or test diffusion outputs.

The computational study accessed only the aggregate workbook and no directly identifying respondent data. The survey materials record online consent, weekly-or-more X use as an eligibility criterion, and administration through an online panel provider. They do not, however, provide in the reproducibility repository a complete fieldwork report containing invitations, response rates, exclusions, final weighting and all transformations from the questionnaire to the workbook. These limits preclude a claim of national representativeness. The aggregate parameterisation supports the stated comparative scenarios, not population inference.

Named strategies copy attributes from traditional media to the experimental source. Engagement covers positive and negative emotion, simple language, sensationalism, entertainment, audiovisual content and hashtags; proximity covers political, geographic and social proximity; informational camouflage covers information quality and links; and credibility camouflage covers source credibility. Combined imitation activates all four groups. Campaign intervals are inclusive. Attributes revert after a temporary campaign, but endorsements accumulated while it was active remain subject to the configured memory rule.

3.4 Outcomes and estimands

Let $N = 500$, $T = 400$, D_t be the number of decisions, F_t decisions attached to false source-period publications, and A_t selections of the experimental source. We report separately

$$R_{AP} = \frac{\sum_t F_t}{NT}, \qquad R_{FD} = \frac{\sum_t F_t}{\sum_t D_t}, \tag{4}$$

$$R_{AD} = \frac{\sum_t A_t}{\sum_t D_t}, \qquad R_P = \frac{\sum_t D_t}{NT}. \tag{5}$$

R_{AP} is the false-repost rate per agent-period, R_{FD} the false share among actual decisions, R_{AD} the experimental-source share among decisions and R_P participation. This resolves the denominator ambiguity under restricted reach or activity. Source-by-source selection totals identify displacement. Cumulative unique reach through period 400 can approach saturation under this population size and decision rule, so we also report unique experimental-source reposters at periods 25, 50, 100 and 200 and first repost time. Times to 25%, 50% and 75% use the retrospectively observed period-400 total as denominator and are descriptive trajectory measures, not prospective diffusion thresholds.

RQ2 uses matched windows. Each treatment is compared with its common-seed control during the campaign and, after removal, in periods 1–25, 26–50 and 51–100. This separates within-campaign efficacy from decay and uses the same distance from removal for every temporary campaign.

3.5 Experimental design and statistical analysis

Table 3 summarises the complete staged design. The original 2,010 runs are retained and reanalysed with corrected denominators and RQ2 windows. The major-revision suite adds 4,680 runs. RQ3 adds realistic recommendation policies, three reach levels, four strategies and receiver-scale sensitivity. RQ4 adds a Morris global sensitivity screen with eight trajectories across ten inputs (Morris, 1991), plus random-fixed-outdegree versus directed small-world networks crossed with three activity levels. A separate set of 360 observational re-executions records RQ3 mechanism counts without changing the 6,690-run primary estimand set. Parameter ranges are listed in Appendix A.

The sensitivity bounds are scenario ranges, not estimated population distributions. They bracket the workbook values while keeping each input behaviourally and numerically valid: memory spans very rapid to persistent evidence; contact count spans 3–10 listened-to contacts; direct reach spans 1–14.7%; WOM magnitude spans half to twice the legacy value; and source probabilities, choice base and attribute contrast vary around their workbook settings. Morris uses a five-level unit cube, step $\Delta = 0.25$, eight seeded random trajectories without trajectory optimisation, and ten stochastic seeds per design point. Treatment-minus-control effects are computed within seed before elementary effects; μ^* and σ then aggregate 80 seed-level effects per factor. Trajectory-cluster bootstrap and leave-one-trajectory-out ranks assess screening stability. Structural cells separately test whether conclusions survive reduced activity and a clustered directed topology.

Table 3: Staged experimental design. Run files are stochastic replications; common-seed treatment and control files are intentionally dependent.

RQ	Conditions	Runs	Main factors
RQ1	6	180	Five source strategies and control
RQ2	19	570	Strategy, start period and duration
RQ3	92	2,760	Reach, strategy, recommendation policy, label quality and receiver scale
RQ4	306	3,180	Memory/connectivity grid, ten-factor Morris screen, activity and topology
Total	423	6,690	Original and major-revision suites

For each condition, treatment minus matched control defines the seed-level effect. We report the mean effect, unadjusted Student-t 95% Monte Carlo interval, paired d_z, baseline mean, absolute percentage-point change and relative change. Holm correction is applied to p values within each RQ–outcome family; intervals are not multiplicity adjusted. Statistical significance is not treated as empirical uncertainty about X. Practical interpretation uses both absolute and relative magnitude and labels intervals crossing zero as imprecise rather than evidence of no effect. For the Morris screen, μ^* ranks mean absolute elementary effects and σ indicates non-linearity or interaction.

Thirty seeds are used for each RQ1–RQ3 condition, 15 for the original RQ4 grid and 10 for each global or structural sensitivity cell. This allocation prioritises broad uncertainty coverage in the computationally expensive sensitivity design. The public condition and effect tables provide each standard error and interval, from which realised Monte Carlo half-width is $(\mathrm{CI}_{high} - \mathrm{CI}_{low})/2$; these quantities do not convey population-sampling uncertainty.

3.6 Software execution and reproducibility

The Java implementation separates model execution from study orchestration, so an individual workbook can still run without the study layer. It was adapted from an earlier, independently published agent-based simulation code line for cross-border e-commerce (Terán et al., 2022, 2025), then materially re-specified for source-mediated news dissemination, source publication states, user evidence, recommendation policies and the present experimental design. The earlier applications establish software lineage, rather than empirical validation of false-news diffusion. The revision suite was executed with ten concurrent Java processes on a MacBook Pro with an Apple M2 Max processor (12 CPU cores: eight performance and four efficiency), 32 GB RAM, macOS 26.6.1 and OpenJDK 25.0.1. The final resumed execution stage began with 2,130 completed runs and produced the remaining 2,550 in 14 h 10 min; this interval is not presented as the elapsed time for the entire computational campaign. The completed/failed counts, environment and software revision are exported with the validated study artefacts.

The open-source repository contains the executable study definition, input workbook, fixed seeds, tests, analysis scripts, processed one-row-per-run data, statistical contrasts, figures and SHA-256 checksums. Raw generated workbooks are excluded because of their size but are reproducible from the published inputs and seeds. The public repository is `https://github.com/pleger/FAKENEWS-ABM-News-Source-Strategies` (Leger et al., 2026a); the submitted manuscript cites the immutable public release `v0.2.0` and its recorded source revision.

4 Scenario analysis

This section answers the four research questions in sequence. Each analysis distinguishes malicious-source audience outcomes from network-wide false-repost outcomes, reports uncertainty from paired seeded runs, and closes with a boxed statement of the principal result.

4.1 RQ1: source-imitation strategies

Figure 3 shows that increasing the experimental source's audience did not automatically increase overall false-news exposure. The control mean was 1.328% of decisions for the experimental source and 11.464% false decisions. Informational and credibility camouflage increased source share by 0.087 pp (95% CI 0.065–0.109) and 0.048 pp (0.020–0.076), respectively. Combined imitation produced the largest audience effect, 0.471 pp (0.399–0.543), a 35.5% relative increase over control, and 17.17 additional unique reposters by period 400 (13.74–20.59).

The corresponding false-decision effects were smaller. Combined imitation added 0.157 pp (−0.017 to 0.331 pp; $d_z = 0.337$; Holm $p = 0.301$), a 1.4% relative change whose interval is imprecise. Proximity reduced the false share by 0.033 pp (−0.055 to −0.012; $d_z = -0.584$; Holm $p = 0.017$) while increasing experimental-source share. Source displacement explains this counterintuitive sign: relative to 200,000 control decisions, proximity added about 79 experimental-source and 526 traditional-source selections while removing about 398 unknown- and 206 mixed-source selections. Combined imitation added about 942 experimental-source selections, primarily displacing the unknown source. Thus RQ1 supports audience capture through imitation but not a one-to-one conversion from source success to false exposure.

Period-400 unique reach is close to its ceiling: control already reaches 459.47 of 500 agents. Earlier horizons are more discriminating. At period 25 the control mean is 274.97 and combined imitation adds 34.07 unique reposters; the corresponding treatment effect declines to 17.17 by period 400 as control catches up. This pattern identifies faster audience acquisition rather than an unbounded long-run reach advantage.

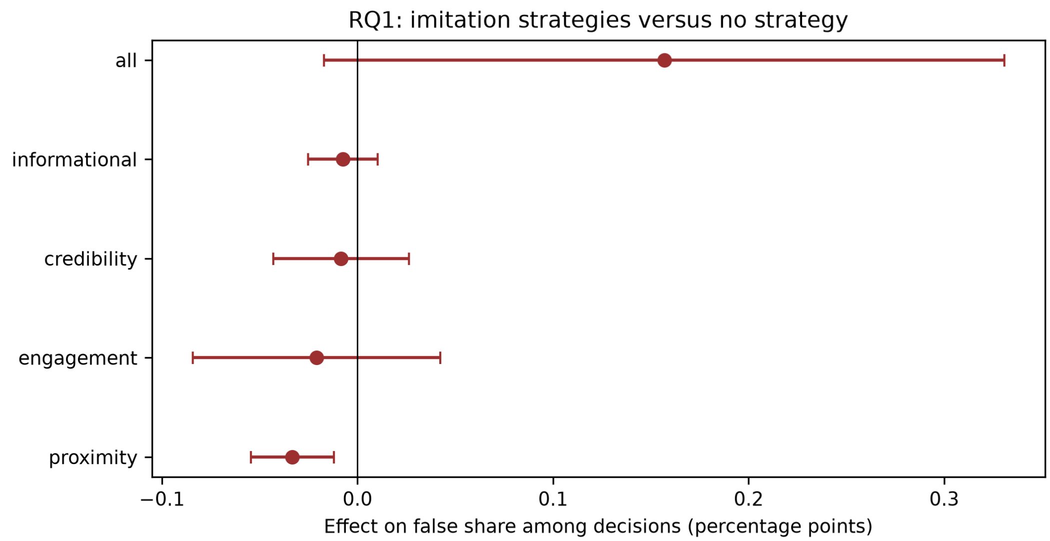

Figure 3: Paired strategy effects on the false share among decisions. Error bars are unadjusted 95% t intervals across 30 matched seeds; Holm correction applies to reported p values, not intervals.

> ***RQ1 key finding.*** *Multidimensional imitation produced the clearest audience gain, but malicious-source popularity and network-wide false-news exposure were not interchangeable outcomes.*

4.2 RQ2: timing, duration, and persistence

Timing and duration changed experimental-source participation during the periods in which a strategy was present. Every temporary and through-period-400 treatment had a positive treatment-window source-share estimate. The largest was the 25-period informational campaign beginning at period 1, 0.498 pp (0.381–0.615), followed by the corresponding credibility campaign, 0.271 pp (0.167–0.375). Full-run averages are smaller because they dilute these short interventions over 400 periods; they are therefore retained only as descriptive summaries, not as the principal timing estimand.

Matched windows reveal dynamics hidden by the full-run mean. During the early 25-period informational campaign, false share among decisions increased by 0.213 pp (0.059–0.367); in periods 1–25 after removal it decreased by 0.212 pp (−0.312 to −0.112), then attenuated to −0.102 pp (−0.230 to 0.027) in post-removal periods 26–50 and −0.010 pp (−0.027 to 0.008) in periods 51–100. The analogous credibility campaign increased the during-window outcome by 0.121 pp (0.007–0.235) and decreased it by 0.140 pp (−0.229 to −0.051) immediately after removal. Other cells were smaller and frequently imprecise. The reversal is consistent with changed source competition and rapidly decaying evidence, not persistent damage continuing in the same direction.

> ***RQ2 key finding.*** *Camouflage increased source share while active. The strongest short campaigns produced an immediate post-removal reversal in false-decision share that largely decayed by 51–100 periods, demonstrating why matched windows are preferable to full-run averages.*

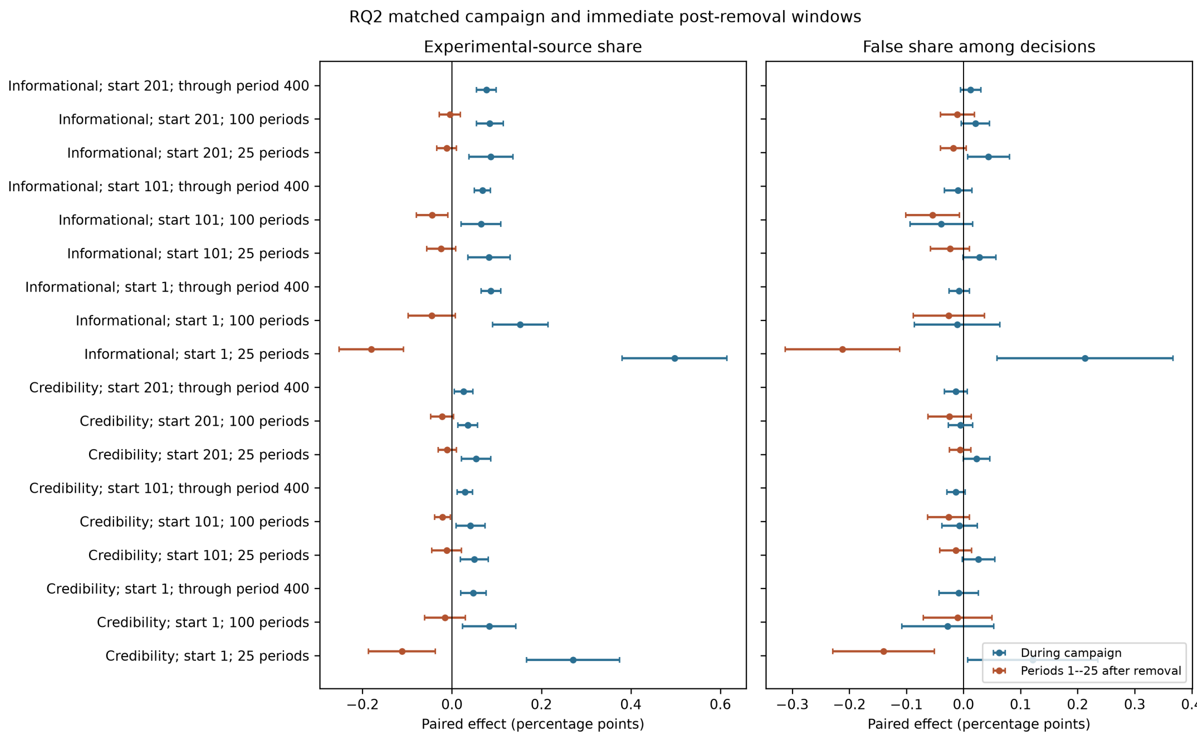

Figure 4: Treatment-window and immediate post-removal effects for RQ2. Points are matched-seed means and bars are unadjusted 95% t intervals across 30 seeds; effects use actual decisions as denominator.

4.3 RQ3: reach restrictions and social recommendations

Recommendation policy materially changed both the baseline outcome and the effect of source imitation. Table 4 reports the combined-strategy effect on false share among decisions at 14.7% direct reach and receiver scale $\lambda = 0.5$. Immediate perfect labels provided the strongest attenuation: the combined effect was 0.149 pp (95% CI 0.111–0.187) from a 9.821% control mean. With recommendations disabled, the corresponding effect was 1.182 pp (1.077–1.287). Discovery-only recommendation increased it to 1.795 pp (1.610–1.979), because discovery expands visibility without providing truth-sensitive evidence.

Imperfect policies produced a qualitatively different result. False-only flagging yielded a 6.457 pp combined effect (6.177–6.737), fast imperfect labels 9.097 pp (8.753–9.441), and delayed imperfect labels 15.131 pp (14.724–15.538). These contrasts do not imply that labelling causes misinformation on X. They show that a modelled recommendation channel that expands source visibility can overwhelm its corrective signal when coverage, classification or timing is degraded.

The separate 360-run diagnostic reproduction makes this mechanism auditable without changing the primary estimand. It reproduced every archived run-level outcome exactly under paired seeds (zero outcome mismatches). In an active receiver-period, each user processed about five contact recommendations. Discovery occurred before label coverage: combined imitation produced 6.011 newly discovered sources per 1,000 receiver-periods under false-only, fast-imperfect and delayed-imperfect policies, versus 3.205 under the perfect-immediate boundary. For false-only flagging, the combined condition penalised 37.66% of covered recommendations, compared with 23.66% for control; the difference reflects the greater frequency with which the combined source was recommended, not a change to the configured label rule. Fast-imperfect and delayed-imperfect labels realised their configured coverage, sensitivity and specificity closely, while delayed evidence delivered 92.18% of scheduled endorsements within the 400-period horizon. Table 5 gives the configuration and observed mechanism rates; the public processed diagnostics retain counts for discovery, coverage, classification, rewards, penalties, duplicates and exact ties.

Reach effects were consequently policy dependent rather than universally monotonic. Under disabled, discovery-only, perfect-immediate and delayed-imperfect policies, the combined false-decision effect was larger at 14.7% than at 1% reach. Under false-only flagging and fast imperfect labels it was largest at 1%, reflecting non-linear changes in both treatment and matched control. Zero direct reach remains an exact absorbing boundary only when recommendation cannot discover the source; it is not a general estimate of account-removal policy.

Receiver magnitude also changed the contrast. Under fast imperfect labels at 14.7% reach, the combined effect decreased from 11.152 pp at $\lambda = 0.25$ to 9.097 pp at $\lambda = 0.5$ and 5.215 pp at $\lambda = 1$. Because magnitude affects treatment and control simultaneously, this pattern is a model interaction rather than evidence that stronger interpersonal influence is intrinsically protective.

Table 4: Combined-imitation effect on false share among decisions at 14.7% direct reach and $\lambda = 0.5$. Effects are percentage-point treatment-minus-control differences across 30 paired seeds; intervals are unadjusted 95% Monte Carlo intervals.

Policy	Control (%)	Effect (pp)	CI low	CI high
Disabled	15.575	1.182	1.077	1.287
Discovery only	15.118	1.795	1.610	1.979
False-only flag	30.400	6.457	6.177	6.737
Perfect and immediate	9.821	0.149	0.111	0.187
Imperfect and fast	28.575	9.097	8.753	9.441
Imperfect and delayed	22.826	15.131	14.724	15.538

Table 5: RQ3 mechanism audit for the combined-imitation diagnostic re-executions. Observed rates are calculated from 30 seeds and 400 periods; discovery is per 1,000 receiver-periods and penalty is per covered recommendation.

Policy	Coverage (%)	Sens. (%)	Spec. (%)	Delay (periods)	Discovery (/1,000)	Penalty (%)
Disabled	–	–	–	–	0.00	–
Discovery only	100.0	100.0	100.0	0	3.56	0.00
False-only flag	100.0	100.0	100.0	0	6.01	37.66
Perfect and immediate	100.0	100.0	100.0	0	3.21	9.37
Imperfect and fast	80.0	90.0	95.0	5	6.01	38.34
Imperfect and delayed	50.0	75.0	90.0	25	6.01	35.76

RQ3 key finding. *Reach governance cannot be evaluated independently of recommendation quality. Perfect and immediate truth signals strongly attenuated strategy effects, whereas incomplete, inaccurate or delayed policies allowed the recommendation channel to amplify them.*

4.4 RQ4: sensitivity and robustness

The original memory–connectivity grid provides a local sensitivity check. Credibility camouflage increased experimental-source share in all twelve cells, with effects from 0.043 to 0.295 pp; only the interval for 10

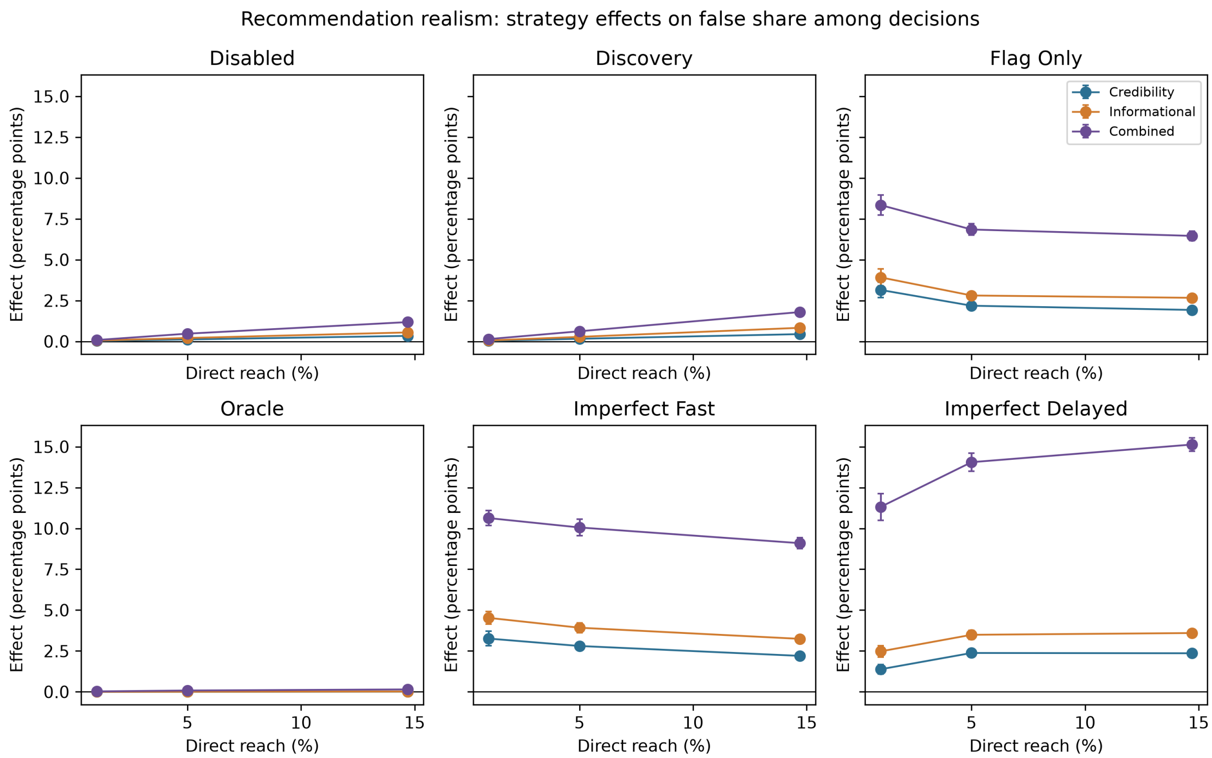

Figure 5: Recommendation-policy and reach sensitivity of strategy effects on false share among decisions. Points are paired means and bars are unadjusted 95% Monte Carlo intervals across 30 matched seeds. The common vertical scale makes policy differences directly comparable.

contacts and half-life 0.33 crossed zero. Its false-decision effect remained positive under the 25-period hard window and five-period half-life, but became smaller under one-period and 0.33-period half-lives. This local result already identifies memory persistence as an important qualification.

The ten-factor Morris screen extends the assessment beyond one-factor grids. For the combined strategy's false-decision effect, memory half-life had the largest mean absolute elementary effect ($\mu^* = 0.806$ pp), followed by target reach (0.599), choice base (0.539), contacts (0.531) and WOM receiver scale (0.374). Their comparatively large σ values indicate non-linearity or interaction rather than stable additive effects. For credibility camouflage, choice base ranked first ($\mu^* = 0.204$ pp); attribute contrast, the unknown-source false probability, WOM scale and reach followed closely. Figure 6 reports both strategy profiles.

Structural sensitivity separates audience robustness from exposure robustness. Experimental-source-share intervals were positive for both credibility and combined imitation in all twelve strategy–activity–topology contrasts. For false share among decisions, combined imitation had a positive interval in five of six cells, whereas credibility camouflage did so in only one. The single imprecise combined cell was the fully active random fixed-outdegree network (0.706 pp, −0.200 to 1.612). Thus the broad audience conclusion survives structural changes, while false-exposure conclusions depend on strategy strength, activity and topology.

> ***RQ4 key finding.*** *Audience capture was robust across local and structural changes. False-exposure effects were less general: memory dominated the combined-strategy screen, interactions were substantial, and weak credibility camouflage seldom produced structurally precise exposure effects.*

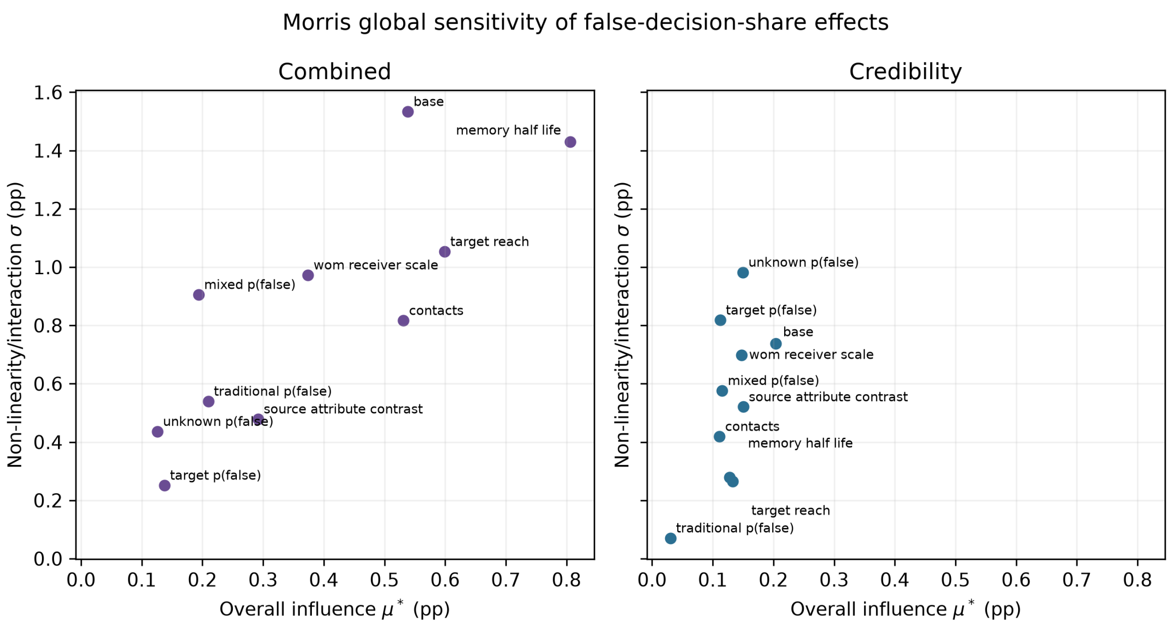

Figure 6: Morris global sensitivity of false-decision-share strategy effects. μ^* measures overall influence and σ indicates non-linearity or interaction across eight trajectories, ten factors and ten stochastic seeds.

5 Verification and validation

This section follows the separation proposed by Sargent (2013): conceptual-model validation, input-data validation, computerised-model verification and operational validation are different claims. It reports the evidence available for each and defines the qualified domain of use. No undocumented expert panel or external acceptance criterion is claimed.

5.1 Computerised-model and analysis verification

The software separates input loading, source and user behaviour, scenario transformation, study orchestration and reporting. Its implementation was adapted from a previously published ABM code line (Terán et al., 2022, 2025), then re-specified and tested for the distinct source-mediated news domain. This lineage provides engineering provenance only: it neither transfers empirical findings from e-commerce nor validates the false-news mechanisms. Invalid memory combinations, probabilities outside $[0, 1]$, unknown strategies, incompatible campaign periods and missing source attributes stop execution. Legacy workbooks remain executable when strategy and explicit source-behaviour sheets are absent, while the study schema separates perceived credibility from objective false-publication probability.

Automated tests cover configuration defaults and fatal validation, hard-window and exponential memory, recommendation outcomes, imperfect-label configuration, strategy resolution, temporary restoration, seeded reproducibility, source publication ordering, activity, topology configuration, output reporting, study planning and backward compatibility. A regression test specifically verifies that source-attribute contrast survives initialization and reset. Analysis tests cover seed pairing, intervals, denominator construction and matched time windows.

The output-validation layer checks unique condition–seed identifiers, completed status, 400 ordered periods, binary publication states, valid decision totals, expected condition counts and bounds for every share. It passed for all 4,680 revision workbooks: 1,860 RQ3 runs across 62 conditions and 2,820 RQ4 runs

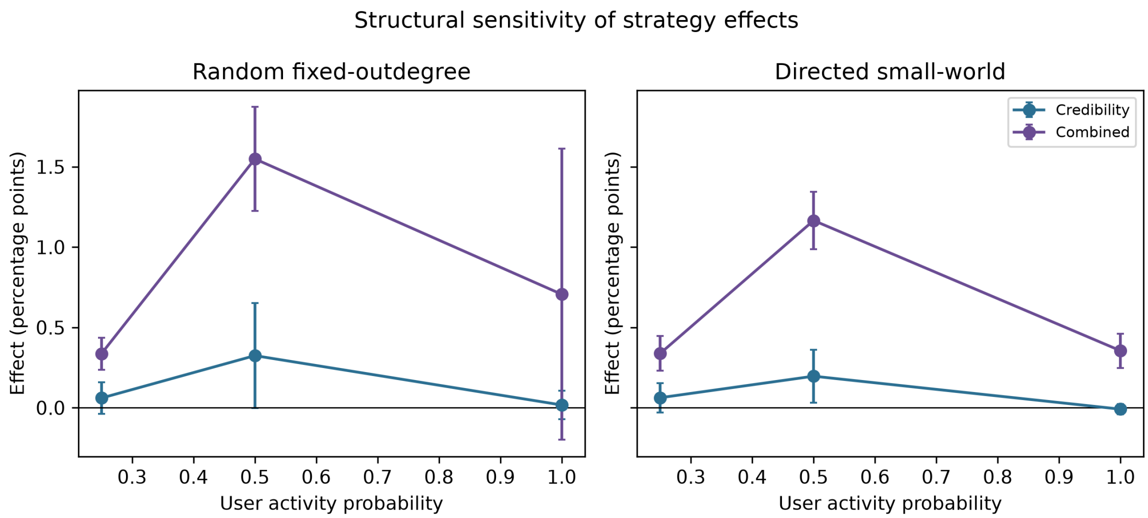

Figure 7: Structural sensitivity of false-decision-share effects across user activity and network topology. Points are paired means and bars are unadjusted 95% Monte Carlo intervals across ten seeds.

across 282 conditions, with no duplicate condition–seed keys. Together with the 2,010 reanalysed runs, the staged design contains 6,690 runs across 423 conditions. A separate 360-run RQ3 diagnostic reproduction passed period-level accounting checks and reproduced every archived primary outcome exactly under the corresponding seed and condition. Processed artefacts include checksums and reconstructable decision totals. These checks demonstrate consistency with the specification; they do not demonstrate correspondence with X.

5.2 Conceptual-model validation

The causal narrative is internally coherent: perceived source attributes affect user evidence and choice; objective publication state is generated independently; recommendation supplies delayed or noisy evidence; and strategy changes presentation without silently changing veracity. Boundary tests support this narrative. An unreachable and undiscoverable source is an absorbing model boundary, not substantive evidence for account removal. Strategy effects can alter source selection without proportional false-exposure changes because false-publication draws remain separate.

Conceptual limitations are material. The model represents four category objects rather than an ecology of competing outlets, one aggregate preference prototype rather than respondent heterogeneity, and no content semantics, homophily, endogenous source optimisation or account replacement. The baseline also requires one decision from every active informed user. Structural and global sensitivity analyses test several consequences of these assumptions but do not make them empirically true.

5.3 Input-data validity

Workbook range, probability-sum, source-coverage and schema checks are enforced. The aggregate survey inputs provide a transparent empirical anchor for perceived source attributes, mean preference weights and initial reach. The available delivery file records 510 complete questionnaires, whereas the model intentionally uses 500 synthetic agents. Consent and institutional-review materials are retained outside the public reproducibility package; the survey-to-workbook transformation, response-rate calculation, exclusions and weighting are not independently reconstructable from that package. The false-publication probabilities

Table 6: Summary of local, global and structural robustness evidence for RQ4.

Analysis	Outcome	Evidence
Local memory/contact grid	Experimental-source share	Credibility-camouflage point estimates were positive in all 12 cells; one interval crossed zero.
Morris global screen	Combined false-decision effect	Memory half-life ranked first ($\mu^* = 0.806$ pp), followed by reach (0.599), choice base (0.539) and contacts (0.531).
Structural sensitivity	Experimental-source share	Both strategies had positive 95% intervals in all 12 activity–topology contrasts.
Structural sensitivity	False-decision share	Combined imitation had positive intervals in five of six cells; credibility camouflage did so in one of six.

are survey-informed modelling inputs rather than audited error rates. Input validity is therefore partial, and national representativeness is not claimed.

5.4 Operational validity and uncertainty analysis

No observed X cascade is reserved for output validation, so the model has not passed predictive operational validation. Directional checks, source-by-source displacement, early-horizon reach, matched post-campaign windows, imperfect recommendation policies and Morris/structural sensitivity provide stronger internal behaviour evidence than the original design, but remain model experiments. The Morris screen ranks influential inputs under stated ranges; it does not estimate their real-world distributions.

The resulting judgement is *fit for comparative mechanism and scenario analysis with explicit caveats*. The model can identify conclusions that survive controlled changes to source strategy, platform reach and uncertain assumptions. It must not be used to predict repost counts on X, infer the causal effect of a current X algorithm, or generalise the aggregate Chilean inputs without further data validation.

6 Discussion and implications

This section interprets the experimental evidence across outcomes, translates each research-question result into a conditional societal implication, and then states the limitations that constrain policy use. The implications concern platform and public-interest decision making; they are not claims that the simulated mechanisms reproduce the current operation of X.

6.1 Distinguishing source success from misinformation exposure

The central finding is that malicious-source success and misinformation exposure are related but distinct outcomes. A source can gain selection share and unique reposters by resembling traditional media, yet the network-wide false-repost share may respond differently as memory, reach, activity, topology and recommendation evidence change. Evaluations of platform policy should therefore report both source-level and content-level indicators. Removing one source metric from a dashboard can conceal displacement or exposure effects elsewhere.

Combined imitation produced the clearest audience gain and a delayed increase in false reposts. This suggests that policies based on a single surface characteristic are vulnerable to multidimensional camouflage.

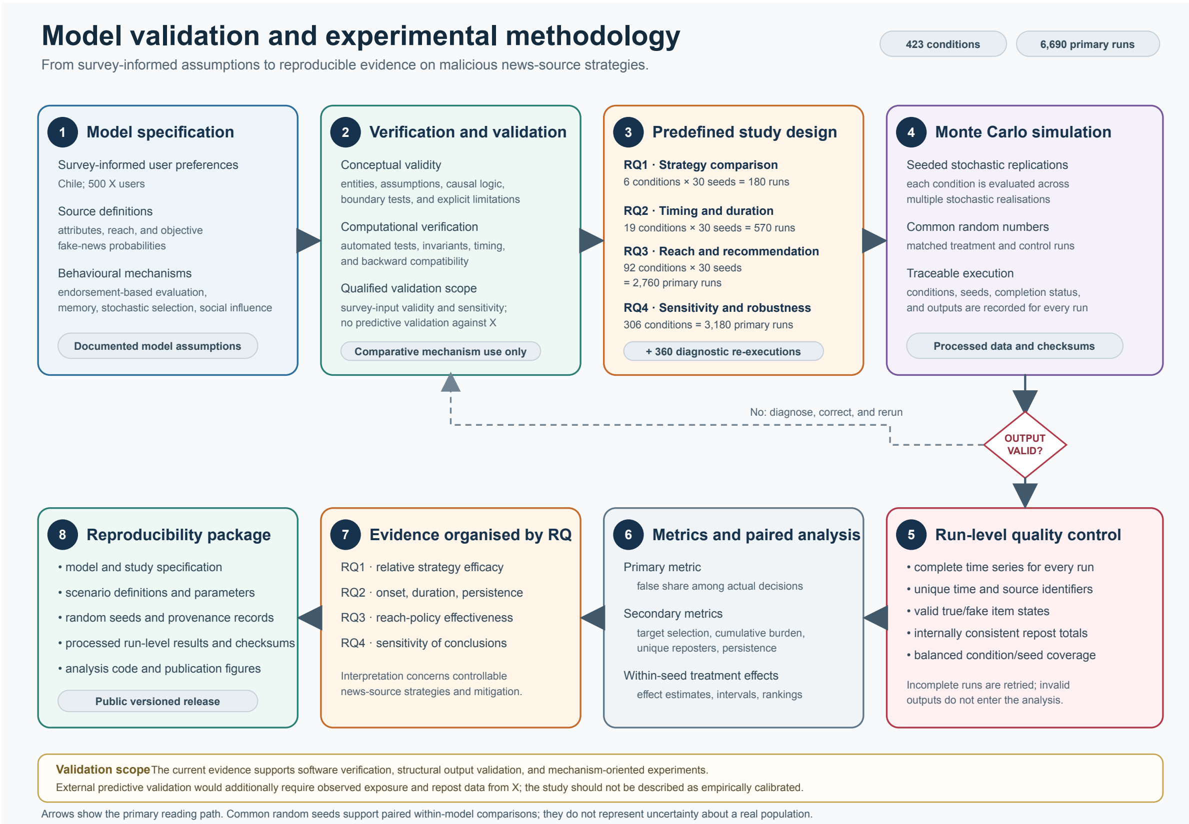

Figure 8: Verification and validation chain. Software and output checks establish implementation consistency; survey-informed inputs, behavioural tests and uncertainty analysis support only a qualified comparative use.

Source assessment should combine provenance, behavioural history, publication-level accuracy and coordination signals instead of relying on appearance alone. The result is compatible with evidence that source-quality judgments can support interventions (Pennycook and Rand, 2019), while extending the question to adaptive source presentation.

6.2 Societal implications of the four scenario analyses

RQ1: multidimensional camouflage. For society, the RQ1 result cautions against treating polished presentation, familiar language, apparent information quality, or a credibility signal as sufficient evidence of reliability. Media-literacy initiatives can teach citizens to examine provenance and publication history, while platforms and fact-checking organisations can combine multiple integrity indicators. Source displacement shows why this should be monitored across the whole information environment: a questionable source can gain users while simultaneous shifts toward or away from other categories change net false exposure. Public reporting should therefore distinguish engagement with a source, actual false redistribution, participation and which alternatives lost attention.

RQ2: timing and duration. RQ2 suggests a social benefit from early integrity monitoring because even a 25-period campaign can change source choice while active. Yet the strongest early campaigns were followed by a short reversal in false-decision share rather than same-direction persistence. A platform evaluation based only on the full-run mean or a distant final window could miss both phases. A proportionate response should therefore track intervention-relative time, distinguish audience retention from content exposure, and avoid assuming either permanent harm or instant recovery after source attributes revert.

RQ3: reach and recommendation governance. Reach and recommendation governance interacted rather than operating as independent controls. Lower reach reduced strategy effects when recommendation was absent, discovery only, perfectly truth aware or delayed and imperfect, but the contrast reversed under false-only flagging and fast imperfect labels because treatment and control followed different non-linear trajectories. Zero reach eliminated the strategy only at an absorbing model boundary where the source could not be discovered. This does not imply that account removal is always desirable or feasible. Real sources can create new accounts, migrate across platforms or exploit indirect sharing, while overly broad restrictions can harm pluralism and legitimate dissent.

Social recommendation requires equally careful interpretation. The perfect-immediate boundary strongly attenuated every strategy, but false-only, imperfect or delayed signals produced much larger effects. The result makes integrity infrastructure part of the policy itself: a visibility intervention that relies on classification must be evaluated together with label coverage, sensitivity, specificity and delay. Public policy should examine auditability, appeals, disparate errors and correction latency, not only the ranking rule that consumes the signal. The receiver-scale experiment further shows that intervention effects can be non-monotonic, so increasing or decreasing recommendation weight without joint scenario testing is not a defensible general prescription.

RQ4: uncertainty and responsible evidence use. Global sensitivity confirms that uncertainty is multidimensional. Memory was the leading influence on the combined strategy, but reach, choice scaling, contacts and WOM magnitude also mattered and interacted. Memory is not a policy variable in the same sense as reach, and the model should not prescribe how citizens ought to remember or evaluate information. Its societal contribution is methodological: decision makers should prefer interventions whose benefits survive credible behavioural and structural assumptions and should disclose when projections depend on uncertain inputs. Audience capture met that standard across activity and topology cells; false exposure did so for the combined strategy in most, but not all, cells and rarely for credibility camouflage alone.

6.3 Limitations and future work

The network is random and small relative to X, users listen to a fixed number of contacts, and all agents use the same decision architecture. Fake-news probability is source-specific but content is not represented semantically. The Chilean survey supplies aggregate inputs without respondent-level uncertainty, detailed sampling documentation, or an external diffusion target. The platform's proprietary ranking, moderation and bot-detection systems are not reconstructed. Results are therefore mechanism-based counterfactuals rather than forecasts. Future work should incorporate empirical networks, user segments, delayed truth assessment, coordinated sources, account replacement, and calibration and validation against observed cascades while preserving the transparent source-behaviour distinction.

7 Conclusions

This section consolidates the answers to the four research questions, identifies the policy conclusions that survive sensitivity analysis, and clarifies the external validation required before predictive use.

This study used an open, survey-informed ABM to evaluate how malicious news sources can imitate trusted-source attributes and how platform-controllable reach and recommendation policies interact with those strategies. Across 6,690 seeded runs in 423 conditions, source camouflage consistently improved audience-related outcomes, with combined imitation producing the largest gain. Campaign timing and duration determined how long the changed source could compete for attention. Immediate perfect truth signals strongly attenuated strategy effects, whereas incomplete, inaccurate or delayed recommendation evidence allowed substantially larger effects. Reach therefore cannot be interpreted independently of the recommendation mechanism that may reveal the source.

The sensitivity experiments provide the qualification needed for policy interpretation. Audience capture remained positive across all structural contrasts. False-exposure effects were less general: memory was the largest influence on combined imitation, several inputs interacted, and credibility camouflage alone seldom produced precise structural exposure effects. Consequently, policy evaluation should not equate malicious-source popularity with misinformation exposure, should use both source- and publication-level measures, and should test recommendation quality and reach governance jointly.

The model is a transparent computational laboratory, not a digital twin of X. Its value lies in controlled comparison, explicit assumptions and reproducible uncertainty analysis. External validation with observed X data is the next step required before predictive or population-level claims can be made.

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Disclosure statement

The authors report no competing interests.

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Ethics statement

The computational study used an existing workbook of aggregate survey-derived parameters and synthetic agents. The modelling and analysis team did not access direct identifiers or contact respondents. The underlying survey protocol was approved by the Ethics Committee of the Universidad Católica del Norte, Coquimbo, Chile (Resolution No. 20/2024).The materials record online informed consent. No chronology beyond these documented facts is asserted.

Data and code availability

The open-source software, study configuration, fixed seeds, processed run-level data and derived analysis artefacts are available in release `v0.2.0` at `https://github.com/pleger/FAKENEWS-ABM-News-Source-Strategies/releases/tag/v0.2.0`. The computational revision is commit `7d5a01d` (Leger et al., 2026a). Generated raw workbooks are not tracked because of their size, but can be reproduced from the published code, input workbook, condition definitions and seeds. Respondent-level survey data are not included.

A Complete ODD model specification

This appendix supplies the self-contained Overview, Design concepts and Details specification of the model following Grimm et al. (2020). It complements the concise methodological account with implementation-level equations, initialization rules and parameter ranges.

A.1 Overview

Purpose and patterns. The purpose is to compare how copied source attributes alter selection of an experimental false-news source and subsequent false-repost exposure under reach and recommendation policies. The model is expected to reproduce directional, not empirical time-series, patterns: greater visibility cannot reduce the set of initially informed agents; a strategy has no direct effect when the source is unreachable and undiscoverable; and longer-lived evidence permits effects to accumulate.

Entities and state variables. Entity types are user agents, four source objects, endorsement events and one scenario manager. User state contains an identifier, a common vector of 14 preference weights (13 source attributes plus word of mouth, WOM), known sources, directed contacts, timestamped endorsements, pending WOM evidence and the most recent selection score. Source state contains an identifier, category, reach, 13 binary attribute distributions, objective false-publication probability, active scenario attributes, publication-state history and a cumulative set of reposter identifiers. Scenario state contains source, target, inclusive interval and copied attribute groups.

Scale. The principal population is 500 users, four source objects and 400 dimensionless periods. Attribute levels are binary. The baseline directed network has five outgoing contacts per user. One publication state is generated by each source in each period.

Process schedule. Initialization loads and validates the workbook, creates sources, creates 500 copies of the aggregate user prototype, samples source visibility, constructs contacts and inserts deterministic period--1 endorsements. Each period then executes scenario changes, source publication draws, pending-label delivery, user activity and choice, reporting, and contact recommendation in that order. Recommendations can affect choice no earlier than the next period plus label delay.

A.2 Design concepts

Basic principles and emergence. Bounded source choice is driven by accumulated attribute-level evidence and contact information. Population shares, exposure and unique reach emerge from repeated local choices; no aggregate trajectory is imposed.

Adaptation, objectives and learning. Users probabilistically choose sources with larger aggregate scores but have no explicit utility maximization or strategic foresight. Choice creates fresh attribute endorsements, so later evaluations are path dependent. Source strategy is exogenous and scheduled; the source does not optimize from feedback.

Sensing and interaction. Users sense only known sources, their endorsement histories and the strongest source recommendation supplied by directed contacts. They do not sense global popularity, network structure or cumulative false-news prevalence. A recommendation can reveal an unknown source.

Heterogeneity. Preference parameters are homogeneous because the only available user record contains survey means. Visibility, network position, activity, sampled attribute experiences, publication exposure, choices and memories generate state heterogeneity. The model does not preserve respondent-level preference distributions or correlations.

Stochasticity. Random draws determine initial visibility, random-network contacts or small-world rewiring, activity, source attribute levels, source choice, source publication state, label coverage and classification errors. A single seeded pseudorandom stream is used within a run. Common seed values improve treatment–control comparability but separate random streams are not used; changes that alter the number or order of draws can therefore weaken common-random-number alignment. Pairing is retained as a design match and this limitation is disclosed.

Collectives, observation and prediction. Contacts form a directed graph but users do not form explicit groups. The reporter records per-period selections, publication states and cumulative unique reposters. Predictions are conditional comparisons within the encoded model, not forecasts of X.

A.3 Details

Initialization. For each source, initial visibility is sampled independently using category reach when reach filtering is active. For every known source and source attribute, the more probable binary level is selected deterministically. If the levels are exactly tied, the first level is retained because the implementation updates the maximum only on a strict increase. With the common weight m_a for attribute a, binary level 1 produces $-m_a/2$ and level 2 produces m_a. Deterministic initialization supplies a common prior-familiarity anchor without consuming treatment-dependent random draws; it removes initial attribute uncertainty and is therefore a modelling limitation. These events are dated -1 and treated as age zero at the first decision. The random fixed-outdegree network samples distinct non-self contacts. The alternative directed small-world network begins with forward ring neighbours and rewires each edge independently with probability 0.1; collisions are resampled, so outdegree remains fixed while indegree varies.

Attribute sampling and endorsement. During interaction, a binary source level $L_{sat} \in \{1, 2\}$ is sampled from the active source distribution. It creates signed value

$$v_{isat} = \begin{cases} -m_a/2, & L_{sat} = 1, \\ m_a, & L_{sat} = 2. \end{cases} \tag{6}$$

The score transform, memory weighting and choice normalization are given in Section 3.2.

Publication behaviour. At the start of each period, source s draws $Z_{st} \sim \text{Bernoulli}(p_s)$. This state is independent of the current attribute realization and is shared by every user selecting the source in that period. Explicit p_s therefore prevents a credibility-copying strategy from changing objective veracity. Legacy workbooks without a source-behaviour sheet instead derive p_s from the current low-credibility probability; that rule is retained only for backward compatibility and is not used by the study workbook.

WOM and labels. Each friend exposes its selected source and current score. For duplicate source recommendations the receiver retains the maximum score, then selects the source with the maximum retained value; exact maxima retain source-identifier iteration order. Given coverage c, a label is sampled independently for that receiver–recommendation event. A false state is classified false with sensitivity s_e; a true state is classified true with specificity s_p. The observed class selects a configured effect $g \in \{-1, 0, 1\}$. Non-zero evidence $g\lambda W$ is dated $t + 1 + d$, where W is common because user preferences are homogeneous. Coverage failure creates no WOM endorsement, although the recommended source has already become known.

Scenario operation. On its first active period, a strategy copies specified attribute distributions from traditional media to the experimental source. Several named groups can be combined. On the period after an inclusive end date, original target attributes are restored. Stored endorsements are not rewritten.

Output calculation. Every active user with at least one known source contributes at most one decision. A selection counts as false when the source-period $Z_{st} = 1$. Unique reach is the number of distinct users who have ever selected each source through the indicated horizon. Formal denominators are given in Section 3.4.

Table 7: Principal parameters and uncertainty ranges. A dash means the value is not varied in that experiment.

Parameter	Principal value	Revision range/levels	Interpretation
Users; periods; source objects	500; 400; 4	fixed	Population, dimensionless horizon and category representatives
Choice base B	1.2	1.10–1.40	Exponential transform of signed endorsements
Memory specification	exponential $h = 0.33$	hard window 25 or $h =$ 0.33–5	Alternative specifications in each run: a hard cutoff, or infinite hard memory with exponential decay
Contacts; friend fraction	5; 1	contacts 3–10	Outdegree is $\lfloor \texttt{CONTACTS} \times \texttt{FRIENDS} \rfloor$
Activity probability	1	0.25, 0.50, 1	Probability that an agent makes its available decision
Topology; rewiring	random; 0.10	random or directed small-world	Fixed outdegree, no self-links/duplicates; variable indegree
WOM receiver scale λ	0.5	0.25, 0.5, 1	Multiplier applied to the common receiver WOM preference
Label delay d	0	0, 5, 25	Additional periods beyond next-period delivery
Coverage c	1	0.5, 0.8, 1	Probability that a recommendation receives a label
Sensitivity; specificity	1; 1	0.75–1; 0.90–1	Classification of false and true source-period states
Source-attribute contrast	1	0.75–1.25	Scales binary probabilities away from 0.5, clipped to $[0, 1]$
Experimental-source reach	14.7‰	0–14.7‰	Probability of initial direct visibility
Traditional false-publication probability	.092	[.05,.15]	Objective source-period probability
Unknown-source false-publication probability	.206	[.15,.30]	Objective source-period probability
Experimental-source false-publication probability	.667	[.55,.80]	Objective source-period probability
Mixed-source false-publication probability	.416	[.30,.55]	Objective source-period probability

Table 8: Survey-informed binary source-attribute inputs. Cells report $P(\text{level }2)$; $P(\text{level }1) = 1 - P(\text{level }2)$. Attribute levels are publication-level stochastic realizations from source-level distributions.

Attribute	Traditional	Unknown	Experimental	Mixed
Entertainment	.884	.899	.667	.820
Geographic proximity	.860	.824	.640	.764
Audiovisual content	.949	.891	.773	.865
Links	.908	.807	.453	.708
Political proximity	.814	.828	.413	.652
Hashtags	.775	.777	.800	.843
Sensationalism	.611	.672	.853	.787
Source credibility	.908	.794	.333	.584
Information quality	.874	.853	.427	.618
Positive emotion	.179	.122	.093	.067
Social proximity	.862	.849	.693	.753
Simple language	.237	.429	.693	.584
Negative emotion	.029	.021	.240	.090

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